

Multiple Pluralization Strategies in Akan: A Split Number Approach

Abstract

Recent work on number has shown that number features may be distributed across different layers of nominal structure. Akan provides a revealing case: multiple number morphemes can occur within a single nominal and differ in their distribution and interpretation. This paper argues that these patterns reflect a split nominal architecture in which number is distributed across three functional heads: a low nominalizing head, a higher renominalizing head associated with humanness, and a Num head. Each plural morpheme realizes a distinct structural position and may make a different semantic contribution. The analysis provides new evidence for split number in Niger-Congo languages.

Keywords: Double pluralization, multiple number marking, Akan, noun class system, prefixation, and circumfixation

1 Introduction

Recent research on nominal syntax has increasingly focused on how number features are represented, particularly within approaches that split the nominal domain into multiple functional projections. One important question in the literature concerns the locus of number: whether number features are introduced on the nominalizing head (n), on a distinct Number Phrase (Num), or distributed across multiple projections. Several studies have proposed that number features may be split structurally, allowing multiple number morphemes to co-occur within a single nominal spine. (Borer 2005; Wiltschko 2008; Harbour 2011; Kramer 2016; Fuchs and van der Wal 2022; Kim and Meng 2022; see also Alexiadou 2019; Dali and Mathieu 2021).

Akan (a Kwa-Niger Congo) provides strong evidence for such a split-number approach. The number morphology is not only overt and productive but also often involves multiple number markers appearing simultaneously within a single nominal. For example, some singular nouns are marked solely by a vowel prefix, while others appear with both a prefix and a suffix (e.g., *V-...-ni*). Plural forms, particularly those denoting humans, show even more complexity. Forms like *a-nua-nom* ‘siblings’ and *a-hene-foɔ*

‘kings’ illustrate how plural prefixes and suffixes can co-occur. In more complex cases, such as *n-namfo-foɔ-nom* ‘friends’ and *a-kwantu-foɔ-nom* ‘travelers’, up to three plural morphemes appear on a single noun.

These data pose challenges for competition-based approaches to morphological realization within Distributed Morphology (DM). For instance, Ofori (2016) analyzes Akan plural markers as Vocabulary Items (VIs) realizing the same [PL] feature under Num. A representative subset of these VIs is given in (1).

- (1) a. /n-/ ↔ [PL] / elsewhere
 b. /a-/ ↔ [PL]
 c. /a-foɔ/ ↔ [PL]
 d. /-foɔ/ ↔ [PL] / _ [+human]
 e. /-nom/ ↔ [PL] / _ [+kinship] (Ofori 2016: 59)

Although this approach is the first theoretical analysis of Akan number morphology and is empirically motivated, it runs into two key problems. First, it predicts that only a single plural marker should appear within a nominal, in line with the Elsewhere Condition.¹ Forms such as *a-nua-nom*, *a-hene-foɔ*, and especially *n-namfo-foɔ-nom* clearly contradict this prediction, since multiple plural morphemes can appear in a single nominal.

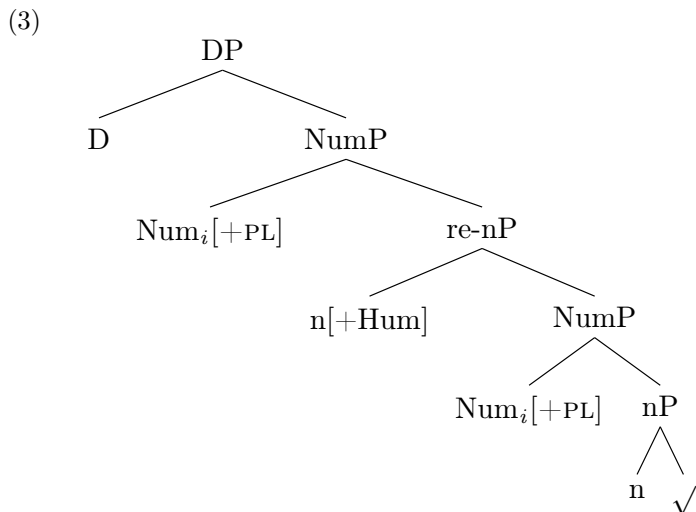
Secondly, the way plural markers appear in Akan is not arbitrary. Their distribution appears to be sensitive to both morphosyntactic and semantic properties of the base noun. For instance, although suffixes like *-foɔ* and *-nom* are limited to [+human] nouns and generally appear alongside an overt plural prefix, these suffixes cannot be said to be mere allomorphic variants. We will return to their structural implications in §3-§6.

This paper develops a split-number analysis of Akan number morphology within a Distributed Morphology framework. We argue that Akan nominals have a layered internal structure, where different number morphemes realize distinct functional projections. In particular, we propose that Akan nominal structure involves at least three relevant heads:

- (2) a. n: categorizes roots as nominals
 b. re-n: introduces [+human]
 c. Num: introduces the [+/-PL] feature

The syntactic structure in (3) shows how plural morphology can be distributed across these heads without violating locality constraints or principles of morphological economy.

¹The Elsewhere Condition, originally formulated by Kiparsky (1973), states that when two competing rules or Vocabulary Items apply to the same input, the more specific rule blocks the more general one.



In our system, the co-occurrence of multiple plural markers is expected. We propose that prefixes such as *a-* and *n-* correspond to a fused spell-out of the lower *n*+Num heads. The suffix *-foɔ* spells out the higher *n* head marked [+human], while suffixes like *-nom* realize the higher Num. The complex form *n-namfo-foɔ-nom* ‘friends’ thus involves three syntactic heads contributing to its plural interpretation, as shown in (4).

- (4) *n-namfo-foɔ-nom*
 PL-friend-PL-PL
 ‘friends’

This analysis accounts not only for the apparent redundancy of plural markers in Akan but also contributes to the broader theoretical debates about the internal structure of the nominal domain. In particular, it shows that number features are syntactically split and can be independently realized across multiple projections (Kramer 2016; Fuchs and van der Wal 2022; Acquaviva 2008; Kim and Meng 2022).

The remainder of this paper is organized as follows. §2 introduces the morphological inventory of singular and plural markers in Akan, including both overtly marked and unmarked forms. §3-§6 presents the theoretical framework and the formal split-number analysis. §7 discusses the semantic interpretation of plural forms. §8 concludes.^{2 3 4}

²Versions of this paper have been presented at Yale Linguistics Department, the 45th Annual Conference of the German Linguistic Society (Deutsche Gesellschaft für Sprachwissenschaft, DGfS), Workshop on Morphology at Princeton, and Association of Contemporary African Linguistics (ACAL-2024). I thank Veneeta Dayal, Jim Wood, Maria Mercedes Pinango, Larry Horn, Natalie Weber, Augustina Owusu, and the above conferences’ audience for their discussions and comments. For judgment and data, I thank Nana Obeng Pewode Andrew, Esther Acheampong, Faustina Boamah, Jonathan Tenkorang, and the other three consultants who prefer to stay anonymous (for Akan data). All errors remain mine.

³The data presented here are drawn from three Akan dialects: Asante, Kwahu, and Akwapim.

⁴Abbreviations used in this paper: ACC = accusative, DET = determiner, FEM = feminine, FUT = future, MASC = masculine, NMLZ=nominalizer, PL = plural, PRF = perfective, PST = past, SG = singular, VI = Vocabulary Item, 1 = first person, 2 = second person, 3 = third person.

2 Number in Akan Nominals

2.1 Singular Nouns

Akan (Kwa, Niger-Congo) makes a clear morphological distinction between singular and plural nouns. Singular nouns are typically marked with vowel prefixes such as *ɔ-*, *o-*, *e-*, *i-*, *ɛ-*, and *a-*, as illustrated in (5) (see Osam 1996; Bodomo and Marfo 2002).

- | | | | |
|-----|--------------------|-------------------|------------------|
| (5) | a. a- konwa | b. o- nipa | c. ɛ- dan |
| | SG-chair | SG-human | SG-house |
| | ‘chair’ | ‘person’ | ‘house’ |

In addition to these overtly marked forms, Akan also has singular nouns that do not carry any explicit morphological marking, as shown in (6).

- | | | | |
|-----|-----------|-----------|----------|
| (6) | a. maame | b. dadewa | c. prako |
| | SG.mother | SG.nail | SG.pig |
| | ‘mother’ | ‘nail’ | ‘pig’ |

A third group of singular nouns is the case of double singular marking, where a prefix (*o-/ɔ-*) combines with a suffix (*-ni*) to form the circumfixal pattern *o-/ɔ-...-ni*. These forms usually denote humans and are often derived from verbs or non-human nouns (see Christaller (1875); Bodomo and Marfo (2002); Adomako (2012)). Examples are given in (7).

- | | | | |
|-----|-------------------------------------|------------------------------|-------------------------------|
| (7) | a. <i>ɔ-</i> kyerɛ-kyerɛ- <i>ni</i> | b. <i>o-</i> sika- <i>ni</i> | c. <i>o-</i> sukuu- <i>ni</i> |
| | SG-teach-RED-NMLZ.SG | SG-rich-NMLZ.SG | SG-school-NMLZ.SG |
| | ‘a teacher’ | ‘a rich person’ | ‘a student’ |

2.2 Plural-Marked Nouns

Plural morphology in Akan also involves both prefixation and circumfixation. Prefixal plurals typically use *a-/n-*, while circumfixal patterns involve combinations of plural prefixes and suffixes in forms such as *a-/n-...-nom* and *a-/n-...-foɔ*. These combined forms are commonly referred to as double plurals (Osam 1996; see also Bodomo and Marfo 2002; Adomako 2012; Ofori 2016).

- | | |
|-----|---|
| (8) | a. Prefixal: <i>a-/n-</i> |
| | b. Circumfixal-1: <i>a-/n-...-nom</i> |
| | c. Circumfixal-2: <i>a-/n-...-foɔ</i> (with <i>ɔ-...-ni</i> as its singular counterpart) |

These plural forms are not freely interchangeable. Prefixal plurals appear with both animate and inanimate nouns, whereas double plurals are restricted to [+human] nouns.

2.2.1 Double and Multiple Marked Plurals

Osam (1996) was the first to dub these plural configurations double plurals. Subsequent work following Osam has generally distinguished two main patterns: *a-/n-...-nom*, primarily associated with kinship nouns, and *a-/n-...-foɔ* (with singular *o-/ɔ-...-ni*), which is more widely attested with human-denoting nouns (Bodomo and Marfo, 2002; Ofori, 2016).

A closer look at the data reveals a more nuanced distribution of plural morphology. As shown in Table 1, whether a noun allows double plural marking depends not only on animacy but also on lexical semantics, base category, and derivational history. The one feature shared by all nouns that permit multiple plural markers is that they denote human referents. Table 1 summarizes these patterns.

Noun Type	Semantic Features	Plural Morphology	Examples	Gloss
A	[+human, +kinship]	<i>n-...-nom, n-...-foɔ</i>	<i>n-kumaa-nom, n-kumaa-foɔ</i>	sisters-in-law
B	[+human, +social]	<i>a-...-nom, a-...-foɔ</i>	<i>n-nan-foɔ, n-nanfo-nom</i>	friends
C	[+human, +kinship]	<i>a-...-nom</i> only	<i>a-gya-nom, n-sew-nom</i>	fathers, in-laws
D	[+human, +derived]	<i>a-...-foɔ</i>	<i>a-sua-foɔ, a-sika-foɔ</i>	students, rich people

Table 1: Distribution of plural morphology by noun type.

From Table 1, we see that Types A and B allow both *-nom* and *-foɔ*. Type A includes canonical kinship terms, such as sisters-in-law, while Type B covers socially defined roles, such as friends and colleagues. This shows that *-nom* is not restricted to your canonical kinship-denoting nouns.⁵

Type C nouns, though also canonical kinship nouns, differ from Type A in that they allow only the suffix *-nom* (e.g., *a-gya-nom, n-sew-nom*) and consistently reject *-foɔ* (e.g., **a-gya-foɔ*). This restriction does not appear to follow a predictable phonological or syntactic pattern (as far as we are aware), but likely indicates the lexical specification of these roots.⁶

Away from kinship terms, Type D nouns are derived from both non-nominal bases, such as verbs (9a) and adjectives, and from nominal, non-human bases (e.g., *sika* ‘money’ (10a) and *sukuu* ‘school’

⁵Another example is *a-kwantu-foɔ-nom* ‘travellers’. The base *kwantu* ‘travel’ is verbal rather than kinship-related. Here, *-foɔ* nominalizes the root as [+human], allowing further pluralization by *-nom* (cf. Type D).

²A reviewer asked whether there are semantic differences between Type A plural doublets. While a full analysis is beyond this paper’s scope, forms with *-nom* (e.g., *a-nua-nom*) tend to resist numeral and quantificational modification, as in (1), whereas *-foɔ* forms (e.g., *a-nua-foɔ*) combine freely with numerals and quantifiers (see Ahenkorah 2025 for details). For example:

1. #*a-nua-nom* mienu ba-a ha
PL-sibling-PL two come-PST LOC
‘Two siblings came here.’
2. *a-nua-foɔ* mienu ba-a ha
PL-sibling-PL two come-PST LOC
‘Two siblings came here.’

⁶Some kinship roots may be lexically specified as nominal and [+human], so the prefix *a-/n-* alone suffices to derive a human-denoting plural form, blocking further morphological material.

(11a)). In all cases, these forms require overt morphology to become human-denoting nouns. The suffix *-foɔ* introduces a [+human] feature, which, when attached to non-nominal bases, also nominalizes the stem; with nominal bases, it derives human-denoting nouns without changing the category. By contrast, the suffix *-nom* lacks this feature and cannot nominalize, so attaching it directly to verbal or non-human bases yields ungrammatical forms (9)–(11).

- | | | |
|---|--|---|
| (9) a. a-kyerɛ-kyerɛ-foɔ
PL-teach-RED-NMLZ.PL
‘teachers’
b. *a-kyerɛ-kyerɛ-nom | (10) a. a-sika-foɔ
PL-rich-NMLZ.PL
‘rich people’
b. *a-sika-nom | (11) a. a-sukuu-foɔ
PL-school-NMLZ.PL
‘students’
b. *a-sukuu-nom |
|---|--|---|

For *-nom* to occur with non-nominal or non-human bases, the stem must first be nominalized and specified as [+human] by *-foɔ*. Once this step has applied, the resulting human-denoting nominal can subsequently combine with *-nom*, as shown in (12), indicating that *-nom* selects for nominal stems that are already specified as [+human].

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|--|---|---|
| (12) a. a-sua-foɔ-nom
PL-learn-NMLZ.PL-PL
‘learners’ | b. a-sugya-foɔ-nom
PL-bachelor-NMLZ.PL-PL
‘bachelors’ | c. a-kwantu-foɔ-nom
PL-travel-NMLZ.PL-PL
‘travellers’ |
|--|---|---|

The co-occurrence of plural markers in (12) supports a non-competitive analysis of Akan plural morphology, which is the approach we adopt in this study, rather than a system based on simple affix competition.⁷

2.2.2 Single-Marked Plurals

Outside the domain of double-marked nouns, Akan typically expresses plurality through a single prefix, either *a-* or *n-*. These prefixal plurals also appear with certain human-denoting nouns, which can be grouped into two categories: (a) derived nominals, where the prefix contributes to a human interpretation and often carries idiosyncratic meanings (13); and (b) inherently human-denoting roots (14).

- | | | |
|-----------------------------|---|------------------------------------|
| (13) a. sew
‘to sharpen’ | b. a-sew
NMLZ.SG-sharpen
‘in-law’ | c. n-sew
PL-in-law
‘in-laws’ |
|-----------------------------|---|------------------------------------|

⁷For a further discussion of interpretive contrasts between double and triple plural forms, see § 6.5.

(14)	a. nipa	b. o-nipa	c. n-nipa
	‘person’	SG-person	PL-person
		‘person’	‘people’

Historically, Osam (1996) notes that prefixation in Akan reflects a robust Proto-Akan noun class system organized primarily around animacy (see also Christaller 1875; Balmer and Grant 1929; Akrofi 1935; Welmers 1971, 1973; Essilfie 1977; Dolphyne 1988; Dolphyne and Dakubu 1988; Osam 1996, 1994). A simplified reconstruction of this system is shown in Table 2.

Class markers	Examples
Class 1 [+animate] = ɔ-/o-	ɔ-pɔnkɔ ‘horse’
Class 2 [+animate] = a-	a-bofra ‘child’
Class 3 [−animate] = i-	i-dan ‘house’
Class 4 [−animate] = ɛ-/e-	ɛ-kɔ ‘war’
Class 5 [+animate] = n-	m-pɔnkɔ ‘horses’
Class 6 [−animate] = a-	a-dan ‘houses’

Table 2: Reconstructed noun class system in Akan (Osam, 1996).

In contemporary Akan, the link between animacy and prefix morphology has weakened, especially in the plural domain. Table 3 summarizes the current distribution of plural prefixes.⁸

Class	Semantic features	Plural morpheme	Examples
Class I	[+animate] [+human]	<i>a-</i>	<i>a</i> -kyaame ‘chief linguists’, <i>a</i> -hene ‘kings’, <i>a</i> -hohoo ‘foreigners’
Class II	[+animate] [+/-human]	<i>n-</i>	<i>n</i> -nipa ‘people’, <i>n</i> -kraman ‘dogs’
Class III	[−animate]	<i>a-/n-</i>	<i>a</i> -dan ‘houses’, <i>n</i> -konwa ‘chairs’, <i>n</i> -taadeɛ ‘dresses’

Table 3: Distribution of plural prefixes in contemporary Akan.

As Table 3 shows, both *a-* and *n-* now appear with animate (Classes I–II) and inanimate nouns (Class III). The question is why is this the case? Part of the explanation lies in the fact that the prefix *a-* has become a productive nominalizer, forming deverbal nouns that denote culturally salient roles, for instance, *a-kyaame* ‘chief linguist,’ is derived from the phrase *kyaa me* ‘greet me.’ The nouns in Class I show similar patterns.

This nominalizing function of *a-* is also evident in a range of deverbal nouns derived from activity verbs, as shown in Table 4.

By contrast, the prefix *n-* appears frequently in deverbal nouns derived from stative or cognitive predicates, producing more abstract nominalizations (Table 5):

⁸The synchronic Classes I–III in Table 3 do not correspond directly to the reconstructed Proto-Akan Classes 1–6 in Table 2, but represent a simplified view of how the plural prefixes *n-* and *a-* are used today.

Activity verb	Deverbal noun
sa ‘to dance’	<i>a</i> -sa ‘dancing’
kenkan ‘to read’	<i>a</i> -kenkan ‘reading’
didi ‘to eat’	<i>a</i> -didi ‘eating’

Table 4: Activity-based deverbal nominals in Akan.

Stative verb	Deverbal noun
kae ‘to remember’	<i>n</i> -kae ‘remembrance’
tease ‘to understand’	<i>n</i> -tease ‘understanding’
sɛ ‘to resemble’	<i>n</i> -sɛ ‘resemblance’

Table 5: Stative-based deverbal nominals in Akan.

2.2.3 Interim Summary

In short, the use of plural prefixes in modern Akan has shifted over time. Today, whether a noun takes *a*- or *n*- depends not only on the animacy of the noun but also on its meaning. In particular, *a*- has taken on a broader role: it no longer simply marks plurality but can also turn verbs into nouns that refer to people. This helps explain why *a*- now appears with certain human-denoting nouns in contexts where older forms of the language would have used *n*-.

2.3 Unmarked nouns

In addition to morphologically marked nouns, Akan also has a class of nouns that show no overt distinction between singular and plural, as illustrated in Table 6.

Plural	Singular
<i>sumiɛ</i> ‘pillows’	<i>sumiɛ</i> ‘pillow’
<i>nwoma</i> ‘books’	<i>nwoma</i> ‘book’
<i>mpa</i> ‘beds’	<i>mpa</i> ‘bed’
<i>bonsu</i> ‘whales’	<i>bonsu</i> ‘whale’

Table 6: Examples of morphologically unmarked nouns in Akan.

These nouns are not randomly distributed. Rather, they tend to cluster into a few semantically coherent categories: (i) body parts, such as *nsa* ‘hand(s)”; (ii) fruits and vegetables, e.g., *ankaa* ‘orange(s)”; and (iii) a heterogeneous set of semi-lexical items, including *mpa* ‘bed’, *mpaboa* ‘slippers’, and *sumiɛ* ‘pillow(s)’. Syntactically, they behave like count nouns: they can combine directly with numerals without requiring a classifier or measure interpretation, as illustrated in (15).⁹

- | | | | | |
|------|----|---------------|----|--------------|
| (15) | a. | sumiɛ mmienu | b. | sumiɛ baako |
| | | pillow(s) two | | pillow one |
| | | ‘Two pillows’ | | ‘One pillow’ |

⁹We thank the reviewer for prompting this clarification. While some variation exists, these unmarked forms generally belong to semantically coherent subclasses rather than being entirely idiosyncratic.

2.4 Summary

Akan has a richly layered and structurally complex system for marking number. Singular nouns are typically marked with vowel prefixes or circumfixes (V-...-*ni/nyi*), while plural nouns are formed either through simple prefixation (*a-/n-*) or, in the case of human-denoting nouns, with double or multiple markers such as *-nom* and *-foɔ*.

The fact that a single noun can carry more than one number marker shows that Akan plural morphology is more than just a set of interchangeable allomorphs. Instead, each marker marks a distinct syntactic projection and contributes to a specific interpretation.

3 Theoretical Assumptions and Proposal

This paper adopts the framework of Distributed Morphology (DM) (Halle and Marantz, 1993, 1994), in which words are built in the syntax from acategorical roots and functional morphemes. In this approach, roots are category-neutral until they combine with functional heads, which assign both syntactic category and interpretive content, after which Vocabulary Insertion (VI) occurs. VI is governed by the Subset Principle (Halle, 1997, 2000), which ensures that a Vocabulary Item can be inserted at a terminal node only if it matches all or a subset of that node’s features.

Following standard DM assumptions (Marantz, 1997; Arad, 2003; Harley, 2014) and building on Kramer (2016), we propose that number features in Akan are split across multiple projections within the nominal spine. This allows us to account for both single- and double-marked plural forms and to show how different morphological configurations correspond to distinct semantic interpretations.

3.1 Overview of the Split Number Hypothesis for Akan

The core idea of our proposal is that Akan number morphology involves multiple syntactic heads, each contributing specific nominal and number-related features. In this approach, there is a distinction between higher and lower projections in the nominal domain, with each layer interacting with both morphology and semantics. The key projections within the nominal spine are as follows:

- **Nominalizing Phrase (nP)**: The lowest projection, which merges directly with the root. This head categorizes the root as a noun and may carry uninterpretable number features.
 - **Renominalizing Phrase (nP)**: A higher nominalizing head that introduces a [+human] feature. This projection is associated with suffixal morphology (e.g., *-foɔ/-ni*) and sits above a NumP. Although formally identical to the lower n, its higher position results in distinct morphological and semantic effects.
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- **Number Phrase(s) (NumP)**: One or more projections that encode interpretable number features (e.g., [+PL]). Depending on the noun, Akan may realize one or two NumPs.

Under standard minimalist assumptions about feature valuation and interpretability (Chomsky et al., 2000; Chomsky, 2000, 2001; Pesetsky and Torrego, 2004), uninterpretable features must be checked against corresponding interpretable features to avoid derivational failure. In Akan, the distribution is as follows:

- NumP hosts interpretable number features (e.g., [+/-PL]).
- nP hosts uninterpretable number features, which are checked by the appropriate Num head.

This architecture explains why multiple overt number markers can appear on a single noun and predicts their patterns of co-occurrence and distribution.

Further evidence comes from interpretive contrasts in downward-entailing contexts. In English and many other languages, plural expressions in such contexts license inclusive (existentially weak) readings; for example, (16) is true even if no individuals satisfy the predicate (Sauerland et al., 2005; Zweig, 2009; de Swart and Farkas, 2010). Akan shows a similar pattern, but this reading is restricted to certain plural forms, most robustly those marked with the form *a-...-nom*, as in (90).

(16) John did not buy toys. [toys = 0]

(17) John a-n-hu a-nua-nom
 John PRF-NEG-see PL-sibling-PL
 ‘John did not see siblings.’ [siblings = 0]

By contrast, other plural forms, such as those marked only with a prefix or those with the suffix *a-...-foɔ*, allow both inclusive and exclusive readings, depending on context (we define the context in detail in §7).

(18) John ε-n-ni **n-konwa** wɔ store mu ho
 John 3SG-NEG-have PL-chair LOC store inside LOC
 ‘John does not have chairs in the store.’
 [True if chairs = 0 or only 1]

(19) Hospital no ε-n-ni a-dɔkota-foɔ
 hospital DEF 3SG.INANIM-NEG-have **PL-doctor-PL**
 ‘The hospital does not have doctors.’
 [True if doctors = 0 or only 1]

These contrasts show that Akan’s plural system is syntactically articulated: the position and type of a plural morpheme directly influence its interpretation. The split-number hypothesis thus captures both the language’s morphological richness and the semantic behavior of plural nouns. A full compositional semantic analysis is developed in § 7.

With this framework in place, we now turn to the specific ingredients needed to elaborate on the split-number hypothesis adopted here, beginning with the syntactic structures that underlie number-marked nouns in Akan.

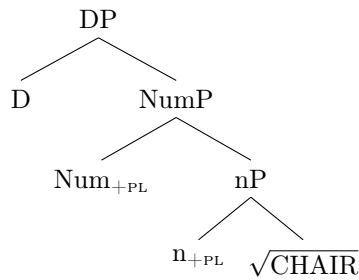
3.2 Overview of Structures for Number-Marked Nouns in Akan

The first ingredient in our analysis is the syntactic structure of number-marked nouns. We propose that Akan exhibits two distinct structures: a simple NumP–nP structure for single-marked forms, and a more complex, recursively embedded NumP–nP structure for double-marked forms. We discuss each structure in turn.

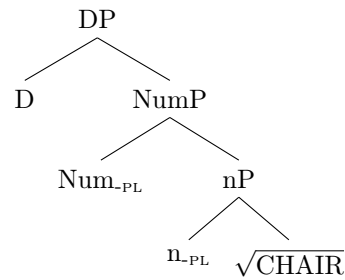
3.2.1 Single-Marked Forms (Structure A)

Nouns with only a single overt number morpheme, typically a prefix, are derived from a structure in which one NumP dominates one nP. This is the structure for both singular and plural prefixal forms, such as *o-nipa* ‘person’, *n-nipa* ‘people’, *a-konwa* ‘chair’, and *n-konwa* ‘chairs’, as shown in (20) and (21).

(20) Plural: *n-konwa*



(21) Singular: *a-konwa*

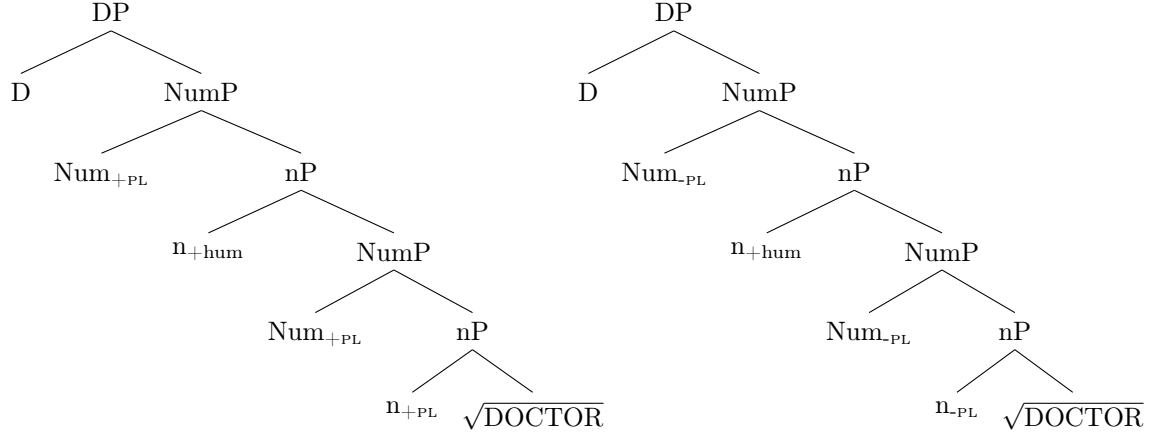


3.2.2 Double-Marked Forms (Structure B)

In contrast, forms such as *a-nua-nom* ‘siblings’, *o-dɔkɔta-ni* ‘doctor’, and *a-dɔkɔta-foɔ* ‘doctors’, which, as noted above, combine both prefixal and suffixal number marking, are derived from a recursive structure in which a higher NumP and nP dominate an inner NumP–nP sequence, as illustrated in (22) and (23). In this structure, the higher nP introduces a [+human] feature and hosts the suffixal morphology, while the inner layer contains the root and the prefix.

(22) Plural: *a-dɔkɔta-foɔ*

(23) Singular: *o-dɔkɔta-ni*



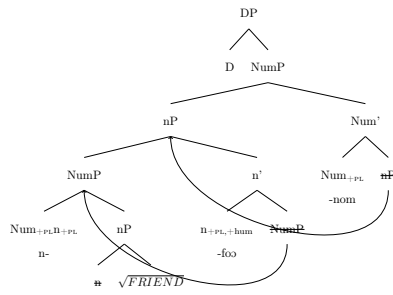
3.3 Deriving Morpheme Order via Movement

While the hierarchical structures presented above capture the relevant syntactic relations, the surface order of prefix–root–suffix in Akan does not follow directly from them. To account for this order, we adopt a movement-based approach to linearization, following the cartographic tradition (e.g., Cinque, 2005). We propose two derivational steps:

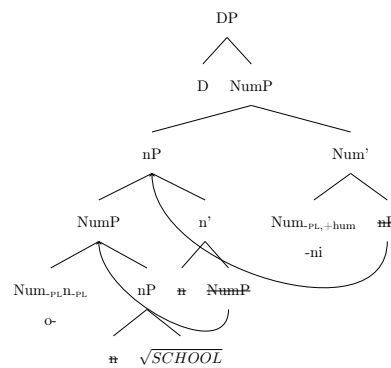
- (a) **Phrasal movement:** The inner NumP moves to Spec,nP, and the resulting nP then raises to Spec,NumP. This step derives the correct surface word order.¹⁰

These movement patterns are illustrated in (24) for *n-namfo-foɔ-nom* ‘friends’ and in (25) for *o-sukuu-ni* ‘student’.

(24)



(25)



- (b) **Head-to-head movement:** This step ensures that the relevant functional heads are adjacent, which is required for post-syntactic fusion¹¹.

¹⁰Similar analyses of affix order based on phrasal movement have been proposed for languages such as Turkish (Kural, 1994), Korean (Kim, 2001), and Hungarian (Bárány, 2015).

¹¹Because we assume head–head movement primarily serves to create adjacency for fusion, we treat it as a post-syntactic operation, even though we illustrate it syntactically in 24 and 25. This approach follows previous work on local dislocation and similar adjacency effects (e.g., Bobaljik (1994, 1995); Embick and Noyer (2001); Myler (2017)), which also treat such operations as post-syntactic. We thank a reviewer for highlighting this point.

3.4 Preliminary Vocabulary Insertion

The final ingredient we consider is morphological realization. We propose that head-to-head movement establishes adjacency between *n* and Num heads, allowing them to fuse. These fused heads are then realized as the prefixes and suffixes *-ni* and its plural counterpart *-foɔ*. Specifically, prefixal morphemes derive from the fusion of the lower *n* and Num heads, whereas suffixal morphemes realize higher functional heads.

The preliminary Vocabulary Insertion rules for these morphemes are given below; they will be further refined in §4.3.1 and §6.2.¹²

Prefixal Vocabulary Items:

(26) Prefixal VI rules:

- a. $n, [+PL] \leftrightarrow a-$ / $-\{\sqrt{\text{INANIMATE}}, \sqrt{OSONO} \text{ “elephant”}, \sqrt{OHEMA} \text{ “queen”}, \dots\}$
- b. $n, [+PL] \leftrightarrow n-$ / $-\{\sqrt{\text{ANIMATE}}, \sqrt{ADAKA} \text{ “box”}, \sqrt{AKONWA} \text{ “chair”}, \dots\}$
- c. $n, [-PL] \leftrightarrow o-$ / $-\sqrt{\text{ANIMATE}}$
- d. $n, [-PL] \leftrightarrow e-$ / $-\sqrt{\text{INANIMATE}}$

Suffixal Vocabulary Items:

(27) Suffixal VI rules:

- a. $\text{Num}[+PL] \leftrightarrow -nom$ / $-\text{nP}_{[+human]}$
- b. $n[+HUM], [+PL] \leftrightarrow -foɔ$
- c. $n[+HUM], [-PL] \leftrightarrow -ni$

3.5 Summary

The syntactic structures, movement derivations, and preliminary Vocabulary Insertion rules together constitute the core ingredients for a split-number analysis of Akan nouns.

With these components in place, we are ready to formally implement the split-number analysis, starting with the prefixes.

4 Analysis of Prefixal Nouns

Building on the core ingredients introduced above, we propose that the prefixes in Akan count nouns realize a fused head, rather than exclusively *n* or Num. In this section, we examine the properties of

¹²The Vocabulary Items proposed here classify roots based on animacy, and also include additional roots that fall outside the animacy-based class.

these prefixes and show why they pattern as portmanteau morphemes.

4.1 Prefixes as the Realization of the Lower n

We start with the lower nP, a part of the nominal structure that has been independently identified as central to both morphophonological and semantic idiosyncrasies. As argued by Marantz (2001), Arad (2003), and Arad (2005), the nominalizing head *n* determines how a root is realized morphologically and can trigger idiosyncratic interpretations.

In Akan, the prefixal morphology exhibits properties similar to *n* heads, making the lower nP an ideal domain from which to begin the analysis of the prefixes. In the discussion that follows, we illustrate how these properties are instantiated by the prefixes.

4.1.1 Selectional Restrictions of the Prefixes

The first property to consider is the selectional restriction property. In addition to marking number, Akan prefixes also perform derivational functions. Like other derivational affixes, they show selectional restrictions: they can target specific syntactic categories or even particular roots.

For instance, the vowel prefixes *o-* and *ɔ-* attach to adjectives to form simulative nominals, as illustrated in (28).

- | | | |
|------|-----------------------------------|--|
| (28) | a. <i>kese</i>
big.SG
'big' | b. <i>ɔ-kese</i>
NMLZ-big
'a big person' |
|------|-----------------------------------|--|

As mentioned earlier, the prefix *a-* attaches to activity verbs to derive gerund-like nominals, as shown in (29).

- | | | |
|------|---|--|
| (29) | a. <i>kenkan</i>
read.INF
'to read' | b. <i>a-kenkan</i>
NMLZ-read
'reading' |
|------|---|--|

Similarly, the nasal prefix *n-* combines with stative verbs (especially cognitive), as illustrated in (30).

- | | | |
|------|--|---|
| (30) | a. <i>kae</i>
remember.INF
'to remember' | b. <i>n-kae</i>
NMLZ-remember
'remembrance' |
|------|--|---|

4.1.2 Idiosyncratic Interpretation

A second diagnostic of n-head morphology is the presence of semantically idiosyncratic or non-compositional meanings. This, too, is observed with Akan prefixes. The prefix *a-*, for instance, yields opaque interpretations with certain roots, as shown in (31).

- (31) a. pam ‘to sew’ → a-pam ‘covenant’
 b. sew ‘to sharpen’ → a-sew ‘in-law’
 c. sɔ ‘to glue’ → a-sɔ ‘hoe’

These meanings are not predictable from the verb meaning alone, showing that the prefix *a-* functions as a category-defining n.

4.1.3 Class Markers

As noted earlier, Akan prefixes also encode nominal class features, particularly animacy, which reflect remnants of the Proto-Akan noun class system (Osam, 1996). For instance, Class 1 animate nouns are marked by *ɔ-* or *o-*, depending on ATR features. Table 7 (repeated from Table 2) provides a summary of this system.

Class markers	Examples
Class 1 [+animate] = <i>ɔ-/o-</i>	<i>ɔ-pɔnkɔ</i> ‘horse’
Class 2 [+animate] = <i>a-</i>	<i>a-bofra</i> ‘child’
Class 3 [−animate] = <i>i-/ɪ-</i>	<i>ɪ-dan</i> ‘house’
Class 4 [−animate] = <i>e-/ɛ-</i>	<i>ɛ-kɔ</i> ‘war’
Class 5 [+animate] = <i>n-</i>	<i>m-pɔnkɔ</i> ‘horses’
Class 6 [−animate] = <i>a-</i>	<i>a-dan</i> ‘houses’

Table 7: Nominal class system in Akan (Osam, 1996).

4.1.4 Category Shift

As shown above, the prefixes derive nouns from verbs (e.g., 29, 30) and adjectives (e.g., 28). In addition, vowel prefixes can shift a noun’s class. For instance, Class 1 prefixes like *ɔ-* can apply to inanimate nouns to yield animate simulative readings, as shown in (32).

- (32) a. *ɛ-boɔ*
 SG.INANIM-stone
 ‘stone’
 b. *ɔ-boɔ*
 SG.ANIM-stone
 ‘a person who is like a stone’

4.1.5 Phonological Sensitivity to Root Properties

Lastly, prefixal choice is also conditioned by phonological processes, especially [ATR] vowel harmony. Akan vowels contrast based on tongue root position: +ATR vowels (e.g., i, e, o, u) versus –ATR vowels (ɪ, ɛ, ɔ, ʊ). Prefix vowels assimilate to the [ATR] value of the root, as shown in (33).

- | | |
|---|---|
| <p>(33) ɔ-pɔnkɔ [-ATR]
 SG-horse
 ‘horse’</p> | <p>(34) ɔ-sukuu-ni [+ATR]
 SG-school-NMLZ.SG
 ‘student’</p> |
|---|---|

Nasal prefixes also undergo homorganic assimilation, surfacing as [m] before bilabials, as shown in (36).

- (36) N-pɔnkɔ → m-pɔnkɔ ‘horses’

In sum, Akan prefixes are sensitive to root-internal phonology and exhibit classic behaviors of category-defining n heads. They attach directly to roots and realize n in the structure [nP n Root].

4.1.6 Section Summary

The evidence presented above shows that Akan prefixes do have category-defining head n properties. But this raises an important question. If the prefixes are n heads, how do we account for the number features often associated with them? The next section turns to this issue.

4.2 Number Features on n

According to Kramer (2016), and more generally within the Distributed Morphology (DM) framework, category-defining n-heads are assumed to carry uninterpretable number features, $u[+PL]$ or $u[-PL]$. These features must be checked in the syntax but cannot themselves value agreement or contribute to the semantic interpretation of number (Marantz, 1997). Consider the structure in (37).

- (37)
- $$\begin{array}{c}
 \text{nP} \\
 \swarrow \quad \searrow \\
 \text{n}_u[+PL] \quad \checkmark
 \end{array}$$

We propose that (37) represents the syntactic structure of pluralia tantum nouns, such as *nhwiren* ‘flowers’ and *apasowo* ‘scissors’. These nouns bear plural morphology but do not trigger number agreement because their n-head carries only uninterpretable number features, which must be checked

¹²Following Appah (2005), the prefix *a-* is underspecified for ATR and compatible with both [+ATR] and [-ATR] roots, as in (35a & 35b).

- | | |
|-------------------------------------|----------------------------------|
| <p>(35) a. a-buɔ ‘stone’ [-ATR]</p> | <p>b. a-bofra ‘child’ [+ATR]</p> |
|-------------------------------------|----------------------------------|

before they can serve as a valued goal during syntactic agreement (Chomsky, 2000, 2001; Pesetsky and Torrego, 2007).

How, then, is agreement realized in nP-only structures, including pluralia tantum, mass nouns, and derived nominals (see §4.4)? This question is particularly relevant because, in Akan, we observe that nouns such as pluralia tantum may trigger either singular or plural agreement on their modifiers, as illustrated in (38). How should this pattern be explained?

- | | |
|--|--|
| <p>(38) a. <i>nhwiren kɛsɛɛ</i>
 flower SG.big
 ‘A big flower’</p> | <p>b. <i>nhwiren a-kɛsɛɛ</i>
 flower PL-big
 ‘Big flowers’</p> |
|--|--|

One possibility we propose is to treat this pattern as an instance of semantic agreement. On this view, agreement on the modifier reflects how the nominal is conceptually construed in context. When *nhwiren* ‘flower(s)’ is conceived as a collective or bouquet-like plurality, i.e. as comprising multiple flower units, plural agreement surfaces. By contrast, singular agreement arises when reference is made to an individual flower unit, such as a single flower or even a salient part of a flower (e.g. a rose bloom).¹³

An alternative analysis is to assume that this variability is a default agreement system in Akan. Under this approach, default agreement is morphologically underspecified, allowing either singular or plural morphology to be realized on modifiers depending on contextual or discourse factors, as illustrated in (38). The core insight shared by both analyses is that these nouns lack a specified number feature; therefore, agreement is not syntactically fixed and must instead be resolved either via semantic agreement or through a syntactic default mechanism.

4.3 Number feature on Num

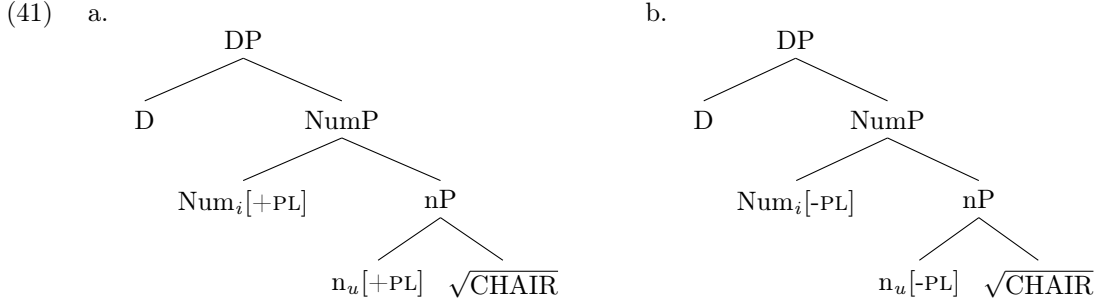
In the literature on number, NumP is generally considered the canonical locus of number features (e.g., Carstens, 1991; Bernstein, 1991). Nominals that project this head are specified for number, giving rise to the singular–plural distinction. In Akan, singular and plural count nouns are distinguished morphologically by prefixes, which not only mark number but also contribute to its semantic interpretation and control agreement, as illustrated in (39) and (40).

- | | |
|---|---|
| <p>(39) a. <i>me tɔ-ɔ a-konwa</i>
 1SG buy-PST SG-chair
 ‘I bought a chair’ [True if one chair was
 bought]</p> | <p>b. <i>me tɔ-ɔ n-konwa</i>
 1SG buy-PST PL-chair
 ‘I bought chairs’ [True if more than one
 chair was bought]</p> |
|---|---|

¹³See Mirković and MacDonald (2013) and Corbett (2023) for discussion of semantic agreement.

- (40) a. **a**-konwa tenten
 SG-chair SG.tall
 ‘A tall chair’
- b. **n**-konwa **n**-tenten
 PL-chair PL-tall
 ‘Tall chairs’

We propose that these count nouns project both *n* and Num: *n* serves as a nominalizing categorizer and bears an uninterpretable number feature, while Num hosts an interpretable $[\pm\text{PL}]$ feature responsible for semantic number interpretation and agreement. The resulting nominal architecture is schematized in (41).



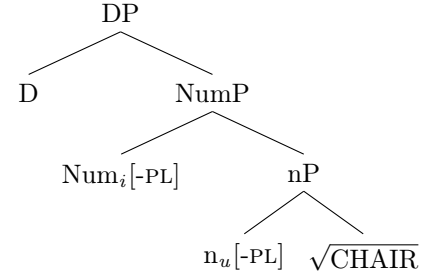
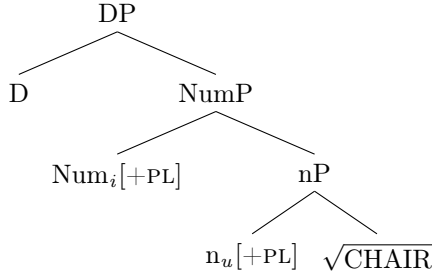
So far, we have seen that Akan prefixes show properties associated with both *n* and Num heads. Interestingly, these two distinct syntactic heads are realized by a single prefix. This raises the question of how separate functional heads can be expressed through a single morphological form.

4.3.1 Fusion of Num and *n*

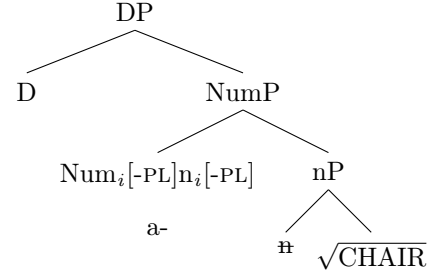
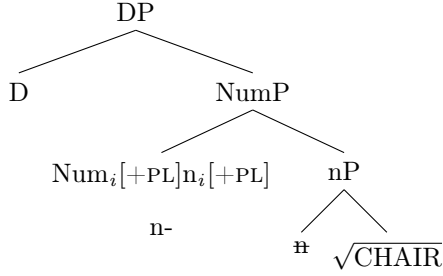
To answer the question above, we propose that this is achieved through the post-syntactic operation Fusion, in which two adjacent heads are merged into a single terminal node (Halle and Marantz, 1993). In this case, Fusion applies only when Num and *n* are linearly adjacent and featurally identical, and this linear adjacency is established via post-syntactic head movement: *n* raises to Num, as illustrated in (43). Once Fusion applies, Vocabulary Insertion targets the fused node, producing a single prefix that simultaneously encodes categorial information and number. The structures before and after Fusion are shown in (42) and (43), respectively, with the relevant Vocabulary Items given in (44). We represent the fused node as *n*, $[\pm\text{PL}]$, as shown in (44).¹⁴

¹⁴Similar head movement has been independently motivated in other nominal systems; see Kramer (2016) for related evidence from Amharic.

(42)



(43)

(44) **Vocabulary Items for the prefixes**

- a. $n, [+PL] \leftrightarrow a-$ / $-\{ \sqrt{\text{INANIMATE}}, \sqrt{OSONO} \text{ “elephant”}, \sqrt{OHEMA} \text{ “queen”}, \dots \}$
- b. $n, [+PL] \leftrightarrow n-$ / $-\{ \sqrt{\text{ANIMATE}}, \sqrt{ADAKA} \text{ “box”}, \sqrt{AKONWA} \text{ “chair”}, \dots \}$
- c. $n, [-PL] \leftrightarrow o-$ / $-\sqrt{\text{ANIMATE}}$
- d. $n, [-PL] \leftrightarrow e-$ / $-\sqrt{\text{INANIMATE}}$ ¹⁵

4.3.2 Interim Summary

This section has shown that, in count noun environments, Akan prefixal morphology realizes both nominalization and number through the post-syntactic Fusion of *n* and *Num*. Under this analysis, *n* carries an uninterpretable number feature, while *Num* introduces an interpretable $[\pm PL]$ specification. The Fusion of these two heads yields a single morpheme that simultaneously reflects both syntactic heads, thereby maintaining morphological economy.

This contrasts with patterns observed in pluralia tantum constructions, where the prefix is associated solely with the nominalizing head, *n*. In these cases, no interpretable number feature is present, and the resulting forms consequently lack canonical number agreement. Other nouns that behave similarly to pluralia tantum cases include mass nouns and derived nominals, which are discussed in the following subsections.

¹⁵The singular prefixes *o-* and *e-* alternate phonologically with $[\text{ɔ}]$ and $[\text{ɛ}]$, respectively, due to ATR vowel harmony, a general property of Akan vowel phonology.

4.4 Derived Nominals

As discussed in § 4.1.1, Akan nominal prefixes are not exclusively markers of number, they also function as nominalizers. One instance where this nominalizing function becomes evident is in the formation of derived nominals. We focus on two types of derived nominals in Akan: (a) similitive nominals and (b) eventive and stative nominals.

Similitive nominals are typically derived from adjectives via the vowel prefix *o-* (or [ɔ]-), as illustrated in (45). Once derived, these nouns behave as ordinary count nouns: they denote singular individuals and trigger singular agreement, as shown in (46).

- | | | |
|------|-------------------------|--|
| (45) | a. <i>kɛsɛ</i>
‘big’ | b. <i>ɔ-kɛsɛ</i>
NMLZ-big
‘a big person’ |
|------|-------------------------|--|

- (46) *ɔ-kɛsɛ tenten no*
NMLZ-big SG.tall DET
‘the big tall person’

We argue that similitive nominals share the same underlying structure as ordinary count nouns, projecting both nP and NumP.

By contrast, eventive (gerundive) nominals formed with the prefix *a-* and stative nominals formed with the nasal prefix *n-*, as in (47) and (48), do not encode number. Like the pluralia tantum nouns discussed in §4.2, these nominals may trigger either singular or plural agreement on modifiers, depending on semantic and discourse context, as illustrated in (49). For instance, a dancing event may be construed as a single extended activity (yielding singular agreement, cf. 47) or as a plurality of salient subevents (yielding plural agreement, cf. 47). As shown in §4.2, such agreement is best analyzed as an instance of semantic agreement. This pattern shows that these nominals lack a NumP within their nominal spine, as schematized in (50).

- | | | | |
|------|---|------|---|
| (47) | a. <i>sa</i>
‘to dance’

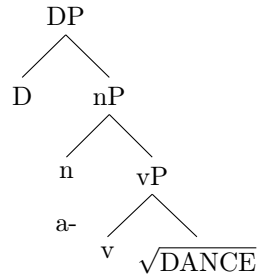
b. <i>a-sa</i>
NMLZ-dance
‘dancing’ | (48) | a. <i>tease</i>
‘to understand’

b. <i>n-tease</i>
NMLZ-understand
‘understanding’ |
|------|---|------|---|

- (49)

a.	a-sa	tenten	nkoa	na	wɔ	b.	a-sa	a-tenten-a-tenten	nkoa
	NMLZ-dance	long.SG	alone	FOC	3PL		NMLZ-dance	PL-tall-RED	alone
	sa-a	yɛ					na	wɔ	nim
	dance-PST	do					FOC	3PL	know
	‘It is only long dancing that they danced.’						‘It is only long dancing that they know.’		

(50) a-sa ‘dancing’



4.5 Mass Nouns

The nasal prefix also appears to occur in mass nouns like *ngu* ‘oil’ or *nsuo* ‘water’. These forms are morphophonologically similar to prefixed plural count nouns but behave syntactically and semantically like mass nouns, as illustrated in (51).

(51) **Mass nouns**

ngu ‘oil’

nsuo ‘water’

nku ‘cream’

Plural count nouns

n-konwa ‘chairs’

m-mofra ‘children’

m-moa ‘animals’

As in English, mass nouns in Akan cannot combine directly with numerals without a measure or container interpretation. In (52), for instance, the phrase is interpreted only in a coerced sense (e.g., ‘two bottles of water’ or ‘two rivers’), and not #two waters.

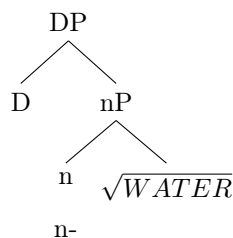
(52) *nsuo* mienu

water two

‘#Two waters’ (interpreted as ‘two bottles of water’ or ‘two rivers’)

Following Borer (2005), we assume that mass nouns also lack a NumP layer. Their structure projects only up to nP, as shown in (53).

(53) *n-suo* ‘water’



A natural question that arises is whether we are dealing with the same prefixes in all the examples discussed so far. Our answer is that these prefixal instances realize four distinct types of the *n* head, and this forms the focus of the next subsection.

4.6 Typological Implications

The distribution of Akan prefixes across count, mass, and derived nominals motivates a more nuanced typology of *n* heads. The relevant distinction we want to highlight is that it is not simply between interpretable and uninterpretable number features on *n*, but between two structural configurations: (i) cases in which *n* participates in Fusion with a higher Num head, and (ii) cases in which no such Num projection is present. This yields a four-way classification, summarized in (54), which reflects the underlying structural differences:

(54) **Types of *n* in Akan:**

- a. *n*_[uPL]: inherent plurals (no Num; *n* bears an uninterpretable [pl] feature)
- b. *n*, [_i+PL]: plural count nouns (a fused *n*+Num head; the interpretable [+pl] originates on Num)
- c. *n*, [_i-PL]: singular count nouns (a fused *n*+Num head; the interpretable [-pl] originates on Num)
- d. *n*: mass nouns and derived nominals (no Num; *n* lacks number features)^{16 17}

5 Unmarked Nouns

Aside from prefixally marked nouns, as noted earlier, Akan also includes a class of nouns that lack overt number morphology yet remain generally number-neutral. This class encompasses body parts

¹⁶It is important to note that interpretable singular or plural features on *n* for count nouns do not originate on *n* itself, but are introduced by Num, which subsequently fuses with the *n* head. By contrast, mass nouns, inherent plurals, and derived nominals do not project a Num head and therefore do not undergo fusion. We thank a reviewer for highlighting this point.

¹⁷The classification of mass nouns and derived nouns under a single *n* head is well supported in the literature (Grimshaw, 1990; Alexiadou et al., 2010; Andreou and Lieber, 2020). These works draw parallels between mass nouns and certain derived nominals—particularly gerundive nominals—arguing that such nominals display mass-like properties.

(e.g., *nsa* ‘hand(s)’) as well as various fruits and vegetables (e.g., *ankaa* ‘orange(s)'). Despite the absence of visible number marking, these nouns behave syntactically like count nouns. In particular, they can combine directly with numerals without triggering a packaging interpretation or requiring a classifier, as illustrated in (55).

- (55) *ankaa mienu*
orange two
‘Two oranges’ (not ‘two bottles/packs of oranges’)

This pattern suggests that these nouns project a NumP headed by a semantically active number feature. Following Borer (2005), we assume that Num introduces the cardinality layer responsible for individuating entities within the nominal domain, enabling both quantification and countability.

Unlike other count nouns, however, these unmarked nouns appear to involve a null Num head. This accounts for their ability to combine directly with numerals. Their structural similarity with morphologically marked count nouns sets them apart from mass nouns, which lack a NumP and therefore require packaging reading when appearing in quantified contexts.

6 Double Number Marking

Having established that the prefixes in Akan are single exponents of a fused *n*+Num head, we can now turn to double number marking. In this system, a noun is marked for number by a combination of a prefix and a suffix. Specifically, plural nouns combine a prefix (*a-* or *n-*) with a suffixal morpheme, either *-nom* or *-foɔ*. Similarly, double singular marking involves a singular vowel prefix (*o-* or *ɔ-*) together with the suffix *-ni*.

6.1 Structural Source of Double Marking

As discussed in §3.2.2, double number marking in Akan arises from a layered nominal structure. In this configuration, a lower NumP introduces the inner number and nominalization, a higher *n*P renominates the structure while introducing a [+human] feature, and a higher NumP contributes the higher number. This yields a nominal spine with two nominalizing heads and two number projections:

- (56) [NumP Num [_{nP} *n*_{+human} [NumP Num [_{nP} *n* √]]]]

Morphologically, this structure is realized in two main ways:

- The lower *n*+Num spells out the prefixes (as discussed in §4.3.1), while the higher *n*+Num spells out the suffix *-foɔ* (and its singular counterpart *-ni*), yielding the form *a-...-foɔ/o-...-ni*.

- The lower n+Num spells out the prefixes, and the higher Num head is overtly realized with the suffix *-nom* giving rise to the configuration *a-...-nom*.

We assume that the lower Num head never surfaces independently but always undergoes Fusion with the adjacent lower n head.

In addition to double number marking, triple number marking is also possible. This occurs when all three heads—lower *n*+Num, higher *n*, and higher Num—are morphologically expressed (see § 6.5).¹⁸

6.2 Fusion of n and Num in the Suffixes

As noted earlier, the suffixes *-ni* (singular) and *-foɔ* (plural) simultaneously encode number and a nominalizing [+human] feature, resulting in human-denoting nominals with interpretable number features. For example, when attached to a non-human noun like *sukuu* ‘school’, they derive human singular and plural counterparts:

(57) sukuu ‘school’

- | | |
|---------------|----------------|
| a. o-sukuu-ni | b. a-sukuu-foɔ |
| SG-school-SG | PL-school-PL |
| ‘Student’ | ‘Students’ |

We analyze these suffixes as Vocabulary Items that realize a complex terminal node, formed via post-syntactic Fusion of a [+human] nominalizing head (n) and a Num head bearing [$\pm$ PL]. Like the prefixes, Fusion is enabled by syntactic head movement: the higher n raises to Num, producing a morphosyntactically adjacent structure visible at PF (as illustrated in 59). At PF, the fused head is realized as a portmanteau suffix: *-fo* for [+PL] and *-ni* for [-PL], while the lower projections yield the prefixes (see §4.3.1).

(58) a.

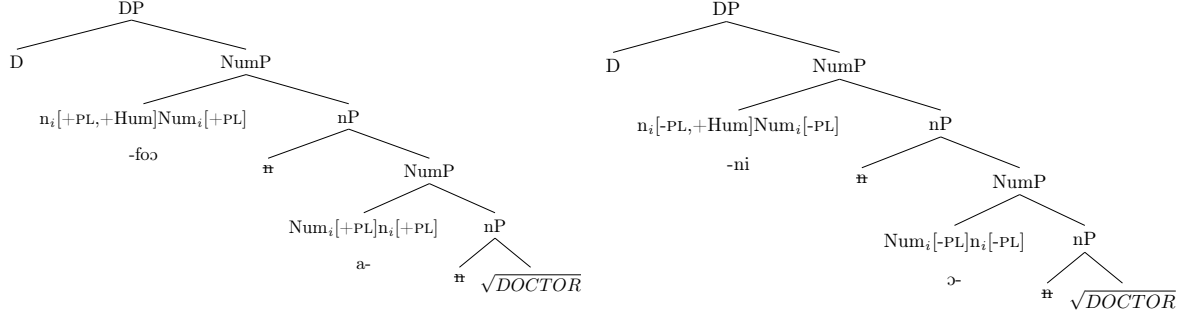
```
graph TD
    DP --> D
    DP --> NumP
    NumP --> Numi["Numi[+PL]"]
    NumP --> nP
    nP --> nu["nu[+PL][+Hum]"]
    nP --> NumP2
    NumP2 --> Numi2["Numi[+PL]ni[+PL]"]
    NumP2 --> nP2
    nP2 --> hash["#"]
    nP2 --> DOCTOR["√DOCTOR"]
```

b.

```
graph TD
    DP --> D
    DP --> NumP
    NumP --> Numi["Numi[-PL]"]
    NumP --> nP
    nP --> nu["nu[-PL][+Hum]"]
    nP --> NumP2
    NumP2 --> Numi2["Numi[-PL]ni[-PL]"]
    NumP2 --> nP2
    nP2 --> hash["#"]
    nP2 --> DOCTOR["√DOCTOR"]
```

¹⁸For comparable cases of double pluralization, see Harbour (2011) on Kiowa, Kramer (2016) on Amharic, Dali and Mathieu (2021) on Arabic, Robinson (2022) on Breton, and Fuchs and van der Wal (2022) on Bantu.

(59)



In terms of agreement, the higher Num projection licenses number concord on modifiers. In double-number marked nouns, where the nominal spine contains two Num heads, an adjective or D head probes for number on the closest Num head, following the locality conditions on Agree proposed in Chomsky (2000, 2001). If the closest Num is valued [SG], the probing head receives singular; if [PL], it receives plural.

After syntactic agreement, post-syntactic Fusion merges $n[+Hum]$ and Num into a single exponent, which is then spelled out as *-ni* (singular) or *-foɔ* (plural), surfacing on adjectives and other modifiers.

- | | | | | | | | | |
|------|----|---------------------------------------|-----------------|-----|----|-------------------------------------|-----------------|-----|
| (60) | a. | a-gyidi-foɔ | a-trenee-foɔ | no | b. | o-gyidi-ni | o-trenee-ni | no |
| | | PL-believe-PL | PL-righteous-PL | DET | | SG-believe-SG | SG-righteous-SG | DET |
| | | a-da | | | | a-da | | |
| | | PRF-sleep | | | | PRF-sleep | | |
| | | ‘The righteous believers are asleep.’ | | | | ‘The righteous believer is asleep.’ | | |

Semantically, these morphemes encode singular and plural interpretations, contributing in a fully compositional manner at LF, as shown in (61).

- | | | | | | | | | |
|------|----|---|--------------|-----|----|---|--------------|-----|
| (61) | a. | Kofi frɛ-ɛ | a-sukuu-foɔ | no | b. | Kofi frɛ-ɛ | o-sukuu-ni | no |
| | | Kofi call-PST | PL-school-PL | DET | | Kofi call-PST | SG-school-SG | DET |
| | | ‘Kofi called the students.’ [two or more] | | | | ‘Kofi called the student.’ [one individual] | | |

In sum, the suffixes *-ni* and *-foɔ* result from the Fusion of a [+human] n head with a Num head bearing $[\pm PL]$, forming a morphologically complex but semantically compositional terminal node. .

6.3 The Suffix *-nom*

As noted earlier, another pattern of double marking involves the suffix *-nom*. Unlike *-foɔ* and *-ni*, however, *-nom* does not derive human-denoting nouns from non-human bases. As a result, inanimate nouns cannot combine directly with *-nom*, as shown in (62b).

- (62) a. n-konwa
 PL-chair
 ‘chairs’
- b. *n-konwa-*nom*
 PL-chair-PL
 Intended: ‘people associated with the chair’ (e.g. chair sellers)

Another important property of *-nom* is that it lacks a singular counterpart. Singular forms are instead expressed either through the prefixes alone or through the circumfixal pattern involving *o-...-ni*.

- (63) a. a-danfo
 SG-friend
 ‘friend’
- b. n-nanfo-*nom*
 PL-friend-PL
 ‘friends’

Importantly, *-nom* may co-occur with *-foɔ*, as shown in (64).

- (64) n-konwa-**foɔ**-*nom*
 PL-chair-NMLZ-PL
 ‘people associated with chairs’ (e.g. chair sellers)

What is particularly striking about *-nom* is that it attaches only to inherently human nouns or to nouns that have already been nominalized as human through *-foɔ*. This restriction is summarized in Table 8.

Noun form	Compatibility with <i>-nom</i>
Ø-maame ‘mothers’	✓maame- <i>nom</i>
n-kraman ‘dogs’	✗*n-kraman- <i>nom</i>
n-konwa ‘chairs’	✗*n-konwa- <i>nom</i>
n-saman ‘ghosts’	✗*n-saman- <i>nom</i>
n-naka-foɔ ‘box sellers’	✓n-naka-foɔ- <i>nom</i>

Table 8: Compatible and non-compatible bases for *-nom*

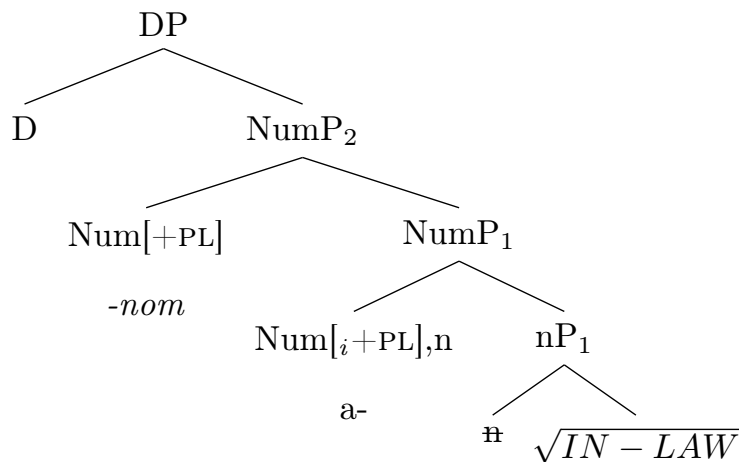
6.3.1 Syntactic Position

I argue that *-nom* realizes the higher Num head. For inherently human nouns, or in cases where the prefix alone is sufficient to yield a human interpretation, the higher nP is not projected. In such cases, the higher Num head attaches directly to the lower NumP.

- (65) n-sew-*nom*
 PL-in-law-PL
 ‘in-laws’

The proposed structure for this pattern is given in (66).

(66)



However, when the base noun is inanimate, an intervening [+human] nP becomes necessary. Without this projection, inanimate nouns cannot combine directly with *-nom*, as shown in (62b), repeated here as (67).

(67)

*n-konwa-*nom*

PL-chair-PL

Intended: ‘people associated with the chair’ (e.g. chair sellers)

Once the noun is first converted into a human-denoting nominal by *-foɔ*, however, *-nom* becomes licensed. This is illustrated by the grammatical form in (68), whose structure is given in (69). The additional *nP2* is therefore required to provide the human-denoting input that *-nom* selects.

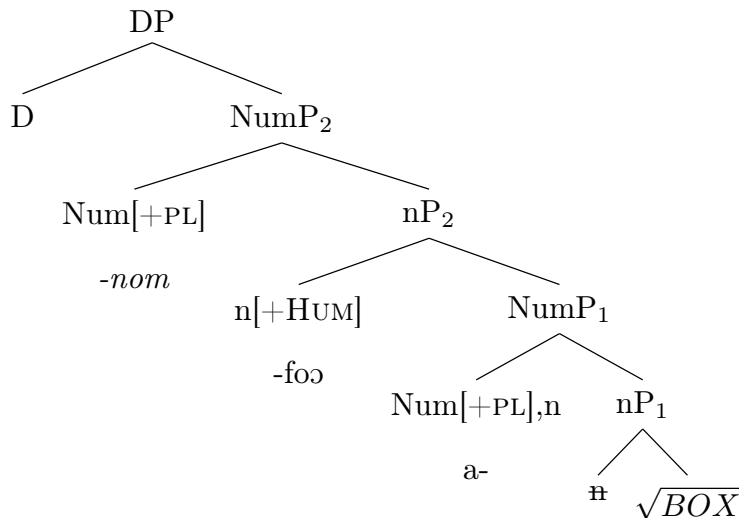
(68)

n-konwa-foɔ-*nom*

PL-chair-PL-PL

‘people associated with the chair’ (e.g. chair sellers)

(69)



This structure also has consequences for agreement. With *-nom*, syntactic agreement is obligatory, particularly on the D head. For example, when the determiner is a demonstrative, it must also surface with *-nom*, as shown in (70).

- (70) n-sew-*nom* wei-nom (*wei)
 PL-in-law-PL DEM-PL (DEM.SG)
 ‘These in-laws’

- (71) a-sew wei (*wei-*nom*)
 SG-in-law DEM.SG (DEM-PL)
 ‘This in-law’

As in §6.2, I assume that the D head carries an unvalued number feature, which is valued by the closest Num head. In (70), this is the highest Num projection, yielding plural agreement.¹⁹

Given the analysis proposed so far, the Vocabulary Item for *-nom* can be stated as in (72):

- (72) **Vocabulary Insertion for *-nom***
 Num[+PL] ↔ *-nom* / _ [+HUM]

This Vocabulary Item covers both structural configurations discussed above. In one configuration, the [+human] feature is introduced by the higher *n* head and is realized by *-foɔ*. In the other, the noun is inherently human and no higher *n* projection is present. In both cases, the plural Num head is in a [+human] environment, and the same Vocabulary Item inserts *-nom*.

In sum, *-ni* and *-foɔ* are fused portmanteau suffixes, whereas *-nom* realizes a higher Num head whose distribution is conditioned by the presence of a [+human] projection.

6.4 Do we need to project all functional heads?

Our analysis so far has focused on cases where all four functional projections are required for double-number marked nouns. But is the projection of all heads always necessary, or can some nouns project only a subset of the functional heads? The evidence suggests a nuanced picture: while some structures do require the full nominal spine, others can be licensed with only a subset of projections.

A particularly case involves nouns with the suffix *-foɔ* that receive a singular interpretation and trigger singular agreement, as shown in (73) and (74).²⁰

¹⁹For the singular counterpart in (71), the human interpretation is derived from the prefix alone, so the higher Num projection is absent. The D head therefore values its number feature from the lower Num projection, which surfaces as null.

²⁰These forms are primarily human nouns, often patient nouns (e.g., *o-yarefo* ‘a sick person’), with occasional exceptions.

- | | |
|--|--|
| <p>(73) a. ɔ-yare-foɔ
SG-sick-NMLZ
'A sick person'</p> | <p>b. ɔ-yare-foɔ kese/ *a-kese no
SG-sick-NMLZ SG.big/ *PL-big DEF
'The big sick person'</p> |
| <p>(74) a. o-didi-foɔ
SG-eat.RED-NMLZ
'A glutton'</p> | <p>b. o-didi-foɔ kese/ *a-kese no
SG-eat.RED-NMLZ SG.big/*PL-big DEF
'A big glutton'</p> |

These data raise the question: how do we account for the co-occurrence of the singular prefix (ɔ- or o-) with *-foɔ*?

Recall that *-foɔ* can serve two distinct functions: nominalizing and plural marking. To account for cases such as (73), we can begin by distinguishing two uses of *-foɔ*:

1. A portmanteau use, in which *-foɔ* realizes both [+human] and plural number and has the singular counterpart *-ni*. This use requires all four functional projections for Fusion, as discussed in §6.2.
2. A nominalizing use, which, for presentational purposes, we represent as *-foɔ_n*. In this use, *-foɔ* realizes only [+human] and does not require a higher NumP. There is no corresponding singular form, as shown by the ungrammatical examples in (75).

- | | |
|--|--|
| <p>(75) a. *ɔ-yare-ni
SG-sick-NMLZ
Int.: 'A sick person'</p> | <p>b. *o-didi-ni
SG-eat.RED-NMLZ
Int.: 'A glutton'</p> |
|--|--|

A direct way of representing this distinction would be to posit the following set of Vocabulary Items for the human suffixes:

- (76) a. $n[+hum], [+pl] \leftrightarrow -foɔ$
b. $n[+hum], [-pl] \leftrightarrow -ni$
c. $n[+hum] \leftrightarrow -foɔ_n$ (elsewhere nominalizer)

Under this analysis, when the higher NumP is absent, *-foɔ_n* spells out a simplex $n[+hum]$ head. When NumP is present and valued [+pl], Num fuses with *n*, and the resulting complex terminal is realized by the portmanteau *-foɔ*.

However, the Vocabulary Items in (76) are more highly specified than may be needed. Under the Subset Principle, the elsewhere Vocabulary Item $n[+HUM] \leftrightarrow -foɔ_n$ can also be inserted when the terminal node bears [+plural], provided that no more specific Vocabulary Item is available. By

contrast, *-ni*, which is specified for [-plural], is the more specific exponent in singular environments and therefore blocks insertion of the elsewhere form. The distinction between the two uses of *-foɔ* therefore need not be encoded by positing two separate Vocabulary Items. The same empirical distribution can be derived with the two Vocabulary Items in (77).

- (77) a. $n[+HUM, -PL] \leftrightarrow -ni$
 b. $n[+HUM] \leftrightarrow -foɔ_n$ (elsewhere)

Why can't the singular suffix *-ni* appear in (75)?²¹ This is explained by the DM Subset Principle, which states that when multiple Vocabulary Items match the features of a node, the most specific item (or the one whose features form a subset of the node's features) is inserted. In the case of $n[+hum]$, *-ni* has an additional feature ([*-pl*]) that is not specified on the node, so it is blocked, whereas *-foɔ_n* matches the node's features exactly and is therefore inserted.

The agreement pattern provides additional support for this analysis. In nouns such as (73), the nominal spine contains only a single number-feature projection, namely the lower Num head. Since the higher Num is absent, the higher *n* head cannot trigger agreement; as a result, the lower Num head serves as the closest available goal. Thus, agreement morphology shows on the modifier as a singular prefix or, in some cases, as a null prefix, as illustrated in (78).

- (78) ɔ-yare-foɔ tenten no
 SG-sick-NMLZ SG.tall DET
 'The tall sick person'

By contrast, when *-foɔ* functions as a portmanteau exponent (e.g., *a-sukuu-foɔ* 'students'), the higher NumP must be projected. Num_i[*+PL*] enforces feature matching across the nominal spine, making singular prefix (*o-*) impossible.

From these observations, we can make the following predictions for the Agree mechanism in Akan. First, there should be no mismatches in the number specifications between the lower and higher projections whenever a higher NumP is present. That is, if a Num head is projected, its [*±pl*] feature must agree with the number feature on the lower Num (see §6.2). This accounts for the absence of combinations such as $n[+pl]$ with Num[*-pl*] (§79).

By contrast, when a higher Num head is absent, and only the higher nP remains, the agreement is determined by the lower Num head (§6.4).

In cases where there is a simple projection with just one nP and no higher Num, agreement is resolved either via default agreement or semantic agreement, as discussed in §4.2.

These predictions are borne out by the attested and unattested combinations illustrated in (79).

²¹We thank a reviewer for drawing our attention to this question

- (79) a. Grammatical Combinations
- i. $n, {}_i[+PL] \dots \text{Num}_i[+PL] \rightarrow a\text{-}\dots\text{-nom}$
 - ii. $n, {}_i[+PL] \dots n[+Hum], {}_i[+PL] \rightarrow a\text{-}\dots\text{-fo}\textcircled{O}$
 - iii. $n, {}_i[+PL] \dots n[+Hum] \dots \text{Num}_i[+PL] \rightarrow a\text{-}\dots\text{-fo}\textcircled{O}\text{-nom}$
 - iv. $n, {}_i[-PL] \dots n[+Hum] \rightarrow \textcircled{O}\text{-}\dots\text{-fo}\textcircled{O}$ (nominalizing exception)
- b. Ungrammatical Combinations
- i. $n, {}_i[-PL] \dots n[+Hum], {}_i[+PL] \rightarrow *$ unattested
 - ii. $n, {}_i[-PL] \dots \text{Num}_i[+PL] \rightarrow *$ unattested
 - iii. $n, {}_i[+PL] \dots n[+Hum], {}_i[-PL] \rightarrow *$ unattested

6.5 Multiple pluralization and group interpretation

Having established the morphosyntactic structure of double number marking in Akan, we now turn to cases of multiple pluralization, in which more than two plural morphemes appear within a single nominal spine. In such cases, plural marking is distributed across these heads: a lower $n+\text{Num}$ head spelled out by prefixes, a higher n head spelled out by *-fo* $\textcircled{O}_n$, and a higher Num head spelled out by *-nom*. Multiple pluralization is most common with human agentive nouns, including terms for students, bachelors, and travelers. More importantly, these triple-marked forms systematically alternate with their double-marked plural counterparts, giving rise to regular plural doublets, as illustrated in (80–82).

- | | |
|--|---|
| <p>(80) a. <i>a-sua-fo</i>$\textcircled{O}\text{-nom}$</p> <p style="padding-left: 20px;">PL-student-PL-PL</p> <p style="padding-left: 20px;">‘students/learners’</p> | <p>b. <i>a-sua-fo</i>$\textcircled{O}$</p> <p style="padding-left: 20px;">PL-student-PL</p> <p style="padding-left: 20px;">‘students/learners’</p> |
| <p>(81) a. <i>a-sugya-fo</i>$\textcircled{O}\text{-nom}$</p> <p style="padding-left: 20px;">PL-bachelor-PL-PL</p> <p style="padding-left: 20px;">‘bachelors’</p> | <p>b. <i>a-sugya-fo</i>$\textcircled{O}$</p> <p style="padding-left: 20px;">PL-bachelor-PL</p> <p style="padding-left: 20px;">‘bachelors’</p> |
| <p>(82) a. <i>a-kwantu-fo</i>$\textcircled{O}\text{-nom}$</p> <p style="padding-left: 20px;">PL-traveler-PL-PL</p> <p style="padding-left: 20px;">‘travelers’</p> | <p>b. <i>a-kwantu-fo</i>$\textcircled{O}$</p> <p style="padding-left: 20px;">PL-traveler-PL</p> <p style="padding-left: 20px;">‘travelers’</p> |

While both forms (*a-...-fo* $\textcircled{O}$ vs. *a-...-fo* $\textcircled{O}\text{-nom}$) denote pluralities, their semantic contributions differ. The simpler forms (e.g., *a-kwantu-fo* $\textcircled{O}$) denote ordinary plural individuals, whereas the more complex

forms (e.g., *a-kwantu-foɔ-nom*) support an interpretation involving a plurality of groups, as shown in (83).²²

- (83) a. A-kwantu-foɔ no ba-a ha anɔpa.
 PL-travel-NMLZ.PL DET come-PST LOC morning
 ‘The travelers arrived here in the morning.’
 (simple plural: one group of travelers)
- b. A-kwantu-foɔ-nom baa ha n-na mmienu a atwa mu.
 PL-travel-NMLZ.PL come-PST LOC PL-day two REL pass LOC
 ‘The various groups of travelers arrived here over the past two days.’
 (multiple groups of travelers)

The form with *-nom* in (83b) introduces an interpretation of pluralities as distinct sets or groups, unlike the simple plural in (83a), which refers to a single collective group.

6.5.1 Group Interpretation in Akan

The discussion so far suggests that at least one of the plural suffixes in Akan gives rise to a group-based interpretation. This raises the question of what exactly is meant by *group* in this context. In English, plural nominals are well known to permit both collective and distributive interpretations, as illustrated in (84).

- (84) The reporters asked the president a question.
- a. Collective reading: the reporters asked a question together.
- b. Distributive reading: each reporter asked a question.

Akan shows a similar ambiguity when the subject is morphologically marked with the plural suffix *-foɔ*, as shown in (85).

- (85) bɔɔl-bɔ-foɔ no a-hyam wɔ parki no so
 ball-play-NMLZ.PL DET PERF-gather LOC park DET LOC
 ‘The footballers have met on the field.’

While both collective and distributive readings are in principle available for plural subjects, the predicate *hyam* ‘gather’ is inherently collective. As a result, (85) is interpreted as describing the footballers acting together as a group. Even if this group is internally structured, for example, if it consists of several football teams, the interpretation at the subject level remains unambiguously collective.

²²I thank a reviewer for raising this point.

The grouping effect becomes more evident when the higher plural suffix *-nom* is added, as illustrated in (86).

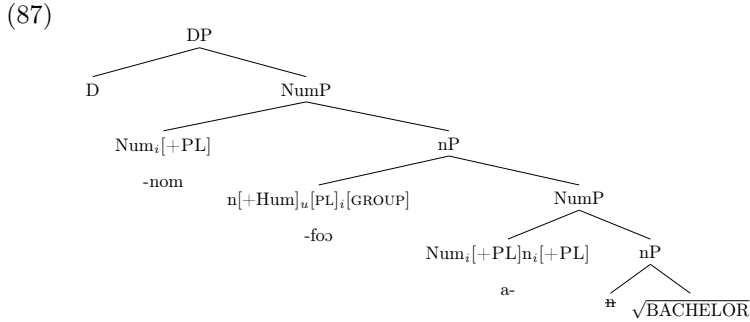
- (86) a-sukuu-foɔ-nom a-hyam-hyam wɔ a-dan a-horow no mu
 PL-school-NMLZ.PL-PL PERF-gather-RED LOC PL-house PL-different DET LOC
 ‘The students have gathered in different rooms.’
 [interpreted as different groups of students in different rooms]

In this example, reduplication of the verb *hyam* gives rise to a distributed-event interpretation, whereby the participants of the subject are associated with distinct spatial subevents. Crucially, the addition of the higher plural suffix *-nom* introduces a higher-order plurality (plurality of sub-groupings), indicating that the subject denotes not just a plurality of individuals, but a plurality of subgroups.²³

In sum, the suffix *-foɔ* introduces a basic level of grouping, while *-nom* stacks an additional layer of plurality over these groups, yielding a structured, higher-order plural interpretation.

6.5.2 Syntactic Structure for Group Interpretation

To capture the interpretive patterns discussed above, we propose that group readings in multiply pluralized nominals arise from a nominal structure in which the higher *n* head introduces an interpretable [GROUP] feature, in addition to a [+human] feature, as shown in (87).



Within this structure, the lower Num+n complex realizes the prefixal plural marker *a-*. The intermediate *n* head introduces the [+human] feature and licenses the suffix *-foɔ*, while the higher Num head spells out as the outer plural suffix *-nom*.

At the level of interpretation, the [GROUP] feature triggers a semantic operation that maps pluralities of individuals onto pluralities of subgroups. The higher Num head then quantifies over these subgroups, resulting in an interpretation in which the nominal denotes a sum of sets/groups rather than a sum of atomic individuals. This compositional structure correctly gives us the observed range of group-level interpretations, such as collections of teams or students observed in the above examples.

²³Reduplication of eventive verbs such as *hyiam* ‘gather’ contributes an event-plural reading, distributing a plural subject across multiple subevents.

7 The semantics of plurality in Akan

We now turn to the semantics of plurality. In the literature on number marking, plurality has been argued either to have a uniform interpretation across languages (see Sauerland et al., 2005; Spector, 2007, among others) or to be semantically ambiguous between an *inclusive* reading (where plural means ‘one or more’) and an *exclusive* reading (where plural means ‘more than one’) (de Swart and Farkas (2010)). In this line of work, the contrast between plural interpretations is commonly derived from the distinction between upward- and downward-entailing environments. Two additional approaches have been proposed in the literature. One derives the contrast from the syntactic position in which number features are introduced (Mathieu, 2014). The other ties plural interpretation to whether the noun is construed as referring to particular entities, in which case quantity is relevant, or to the kind or type denoted by the noun, in which case quantity is irrelevant (Grimm, 2013), and by extension (Martí, 2020).

Cross-linguistically, different plural morphology does not always give rise to semantic contrasts. In some languages, such as Amharic, plural morphemes have been argued to be largely synonymous in their semantic contribution (Kramer, 2016). Akan, however, shows a more nuanced pattern. While plural marking in Akan behaves uniformly in certain environments, it shows systematic variation between inclusive and exclusive readings in others, particularly in downward-entailing contexts such as negation, conditionals, and questions. The availability of one reading over the other depends on the discourse context.

To account for inclusive and exclusive interpretations of Akan plurals, we adopt a hybrid approach. We posit distinct structural positions for the different plural morphemes (as we have been doing so far), and, for those forms that exhibit interpretive flexibility, we follow Grimm (2013) in arguing that plural nouns give rise to exclusive readings in contexts where quantity is relevant and to inclusive readings where quantity is not at issue. This allows us to capture both the morphosyntactic asymmetries among Akan plural markers and their context-sensitive semantic behavior²⁴.

7.1 Exclusive vs. Inclusive Plural Readings in Akan

Plural expressions in Akan can yield either exclusive or inclusive readings depending on the syntactic and semantic environment. In upward-entailing contexts, all plural forms in Akan give rise to an exclusive interpretation, that is, a ‘more than one’ reading (cf. Sauerland et al., 2005; Zweig, 2009; de Swart and Farkas, 2010; Križ, 2017; Martí, 2020; Sudo, 2023 for comparable English data). This is illustrated in (88).

²⁴It is important to note that the semantic differences among Akan plural morphemes extend beyond the inclusive–exclusive distinction alone. As argued in detail in Ahenkorah (2025), these morphemes differ in their underlying semantic composition and in the ways they interact with nominal structure.

- (88) a. Kofi wɔ̃ n-konwa wɔ̃ ne store mu hɔ̃
 Kofi have PL-chair LOC.BE POSS store inside LOC
 ‘Kofi has chairs in his store.’ [more than one chair]
- b. Kofi somma-a a-sukuu-foɔ kɔ̃ɔ ne store mu hɔ̃
 Kofi send-PST PL-student-PL go POSS store inside LOC
 ‘Kofi sent students to his store.’ [more than one student]
- c. Kofi somma-a a-nua-nom kɔ̃ɔ ne store mu hɔ̃
 Kofi send-PST PL-sibling-PL go POSS store inside LOC
 ‘Kofi sent siblings to his store.’ [more than one sibling]

In downward-entailing environments such as negation, conditionals, and questions, plural interpretation in Akan becomes more nuanced and varies systematically with the type of plural morphology involved.

In English, plural nouns under negation typically allow an inclusive interpretation: a sentence such as *John did not buy toys* is true even if John bought none (Sauerland et al., 2005; Zweig, 2009; de Swart and Farkas, 2010). Akan shows a similar pattern, but crucially, this inclusive reading is not uniformly available across all plural forms. It instead appears to be sensitive to the choice of plural morpheme. In particular, the circumfixal plural *a-...-nom* consistently licenses an inclusive (zero) interpretation under negation.

- (89) John did not buy toys. [number of toys bought = 0]
- (90) John ɛ-n-ni a-nua-nom
 John 3SG-NEG-have sibling-PL
 ‘John does not have siblings.’ [number of siblings = 0]

By contrast, other plural forms in Akan, such as prefix-only plurals marked with *a-/n-*, or double-marked plurals of the form *a-/n-...-foɔ*, show greater interpretive flexibility. In downward-entailing contexts, these forms allow both inclusive and exclusive readings, with the intended interpretation determined by discourse context.

Consider the following scenario (**Context A**). John sells chairs. Kofi wants to buy chairs for a party and asks Ama to check whether John has some. John, however, has only one chair. In this context, Ama can report (91).

- (91) John ɛ-n-ni n-konwa wɔ̃ store mu hɔ̃
 John 3SG-NEG-have PL-chair LOC.BE store inside LOC
 ‘John does not have chairs in the store.’

‘If John has chairs, we will have the party.’

The same pattern emerges in Context D with polar questions. When speakers were asked (95) in the competition scenario, they responded negatively but clarified that John does have one sibling, as shown in (96). This indicates that the negative response targets a ‘more than one’ interpretation. When the question was presented without context, however, speakers allowed a ‘one or more’ reading and responded affirmatively.

- (95) John wɔ̌ a-nua-foɔ̌ aa wo be so ne mu anaa
John have PL-sibling-PL COND 3PL FUT hold 3SG.POSS inside QPRT
‘Does John have siblings to help him?’

- (96) Daabi, baako pe na ɔ̌-wɔ̌
no, one only FOC 3SG-have
‘No, he only has one.’

The same contrast was consistently observed across the chair and doctor examples.

7.2 Structural Source of Plural Readings: #P vs. DivP

How should these contrasts be accounted for? We propose that they result from a combination of structural differences in nominal syntax and the interpretive system proposed by Grimm (2013), under which plural nouns yield exclusive readings in contexts where quantity is relevant, and inclusive readings where quantity is not at issue.

On the structural side, we adopt the framework of Mathieu (2014), building on Borer (2005), in which they distinguish between a counting head (#P) and a division head (DivP), and argue that this distinction correlates with exclusive and inclusive plural readings, respectively. Following this approach, we argue that the inclusive–exclusive contrast in Akan plurality follows from the structural position of plural morphology: exclusive plurals arise when plurality is encoded in #P, while inclusive plurals arise when plurality is encoded in DivP.

It is worth mentioning that although our syntactic structures so far have referred only to NumP, we assume that NumP may be interpreted either as DivP or as #P. Under this view, plural morphemes differ in how they map onto these projections.

We have proposed that the plural prefixes *a-/n-* and the suffix *-foɔ̌* are nominalizing heads (n) that lack inherent number features and therefore require valuation by a higher Num head. By contrast, the suffix *-nom* instantiates Num itself and carries an inherent, interpretable number feature, in a manner comparable to the English plural suffix *-s*. Given arguments that English *-s* has an underlying inclusive semantics and has been proposed to project DivP (Borer, 2005), we propose that *-nom* likewise realizes

Div. This accounts for the fact that *-nom*-marked nouns obligatorily permit inclusive readings under negation.

By contrast, nouns marked with *-foɔ* or with plural prefixes exhibit variability between inclusive and exclusive interpretations. This flexibility follows if the NumP associated with these forms can be interpreted either as DivP or as #P. The selection of one interpretation over the other is dependent on whether quantity is discourse-relevant. In upward-entailing contexts, where information about multiplicity or cardinality is at issue (how many did he buy?), an exclusive reading is preferred. The same pattern is observed in conditionals, negations, and questions, as illustrated in Contexts B, C, and D. In all these contexts, the question under discussion is always “how many?”, making the quantity of chairs, siblings, or doctors central to the discourse. In such cases, an exclusive interpretation is favored.

In contrast, in out-of-the-blue contexts or in contexts with no prior information, quantity becomes less salient or irrelevant, and the utterance targets the general type denoted by the noun (e.g., *chair*, *sibling*, or *doctor*). In this case, inclusive readings become available. This suggests that inclusive plurality arises when NumP is interpreted as DivP in the absence of contextual pressure to resolve cardinality.

Another alternative view on why *-nom* allows only inclusive readings, aside from the structural position view, is the argument that the semantics of *-nom* encode maximality (Ahenkorah (2025)). This property explains why *-nom*-marked plurals can receive definite interpretations as bare arguments, a possibility that is unavailable for *-foɔ*-marked nouns. For example, consider a context involving four bachelors and four married individuals, where only the bachelors have eaten. In this scenario, only (97) is felicitous, while (98) yields a weaker, existential interpretation.

- (97) a-sugya-foɔ-nom a-di-di
 PL-bachelor-NLMZ.PL-PL PST-eat-RED
 ‘The bachelors have eaten.’
- (98) a-sugya-foɔ a-di-di
 PL-bachelor-NLMZ.PL PST-eat-RED
 ‘Bachelors (some) have eaten.’

When negation is added to (97), the result is a zero (none) reading, yielding an obligatorily inclusive interpretation.

To wrap it all up, we are arguing that plural morphology is not in semantic competition, just as it is not in syntactic composition: plural morphemes project different syntactic positions and offer different semantic implications.

8 Conclusion

This paper has provided a detailed analysis of number marking in Akan, offering evidence that number morphology in the language cannot be reduced to a competition between morphemes for a single syntactic position. Instead, the data point to a more structured nominal spine in which number features are distributed across three distinct functional projections within the nominal spine: a low nominalizing head (n), a higher renominalizing head associated with humanness and agentivity, and a higher Number Phrase (Num). This architecture captures the co-occurrence of plural prefixes and suffixes and their semantic and syntactic distribution across different nominal classes.

We argued that the plural prefixes in Akan are best analyzed as the realization of a fused [Num+n] complex, with the number feature expressed directly on the nominalizing head. Human-denoting nouns, in contrast, involve an additional functional projection, a renominalizing phrase (nP_{re})—that introduces a [+human] feature and optionally carries its own number specification. The suffixal plural markers, such as *-ni* and *-foɔ*, are the morphological reflexes of number features on this higher nP layer. This layered configuration accounts for double and multiple plural marking patterns without violating locality constraints or the Elsewhere Condition, and it makes precise predictions about the distributional restrictions of different morphological exponents.

These findings offer new empirical support for split number hypotheses in the morphosyntax of nominal structure (cf. Kramer 2016; Fuchs and van der Wal 2022; Acquaviva 2008; Kim and Meng 2022), while also expanding the empirical domain of such analyses to Kwa languages and largely to Niger-Congo. The Akan data show that number marking can be distributed across multiple nominal projections in semantically and morphosyntactically principled ways. In particular, the interaction between number features and animacy-related features such as humanness provides a window into how nominal hierarchies and functional structure are encoded cross-linguistically. The morphosyntactic layering of number in Akan thus serves as a robust testing ground for theories of nominal structure and the interaction of number, animacy, and morphophonology. The split plural architecture proposed here lays the groundwork for further typological and theoretical inquiry into how languages encode fine-grained number distinctions through multiple functional heads.

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